

Continuity modulus of the compact-window Weil form

PRIME.RDAGGER.MODULUS_UPGRADE.01 — round 474

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Abstract

Foundation F2b of the r469 judgment, at fixed support length L . The Guinand–Weil form of r461/r471 is L^2 -Lipschitz in the prime and pole channels with explicit $C_P(L)$, $C_\Pi(L)$, and the piecewise-linear interpolant of an H^2 function obeys the elementary elementwise sine-series Jackson bounds δ^2/π^2 and δ/π . Combined with the r471 grid floor $R_{\text{grid}} = \mu/\delta$, this yields a combination implication on the first Dirichlet H^2 -ball. At every sealed mesh, including the Yoshida/Bombieri point $L = 0.3$, the proved ω_L exceeds R_{grid} (SHORT). The quantitative upgrade $\text{GRID_CERTIFIED} \rightarrow \lambda_*(L) \geq 0$ is STUCK: the modulus is too crude. Lemmas are PROVED. Verdict PARTIAL. No infinitely-many- k statement. No RH claim.

Status. PARTIAL. No promotion of an RH claim.

Claim boundary. Nothing in this note is $\lambda_*(L) \geq 0$ on the r461 L^2 class, nor on any H^2 -ball. The combination theorem is an implication with an empty antecedent on the sealed schedule. The archimedean increment bound is a valid enclosure, not a sharp modulus. No L^* claim, no $R^\dagger$ claim, no RH claim, no anti-RH claim.

1 Object

After r461,

$$\lambda_*(L) = \inf_{\substack{0 \neq h \in L^2 \\ \text{supp } h \subset [-L, L]}} \frac{Q_W(h)}{\|h\|_2^2}, \quad Q_W = A - P + \Pi,$$

with $g = h * \tilde{h}$ and the r471 lag formulae. The form A is not L^2 -bounded (symbol $\sim \log |\xi|$). The working comparison class is therefore the first Dirichlet H^2 -ball

$$\sigma_2(L) = \left(\frac{\pi}{2L}\right)^2 = \frac{\|h''\|_2}{\|h\|_2} \quad \text{for } h(x) = \cos(\pi x/(2L)).$$

Scramble-sensitive: P and $S_{\text{eff}}(L)$ use the literal nodes $\log n \leq 2L$.

2 Proved constants

Lemma 1 (Prime channel). $|P(g) - P(g')| \leq 2S_{\text{eff}}(L) \|g - g'\|_\infty$, where $S_{\text{eff}}(L) = \sum_{n \leq e^{2L}} \Lambda(n)/\sqrt{n}$ is a finite exact interval sum. At $L = 0.3$ one has $2L < \log 2$, hence $S_{\text{eff}} = 0$.

Lemma 2 (Pole channel). $|\Pi(g) - \Pi(g')| \leq 8 \sinh(L) \|g - g'\|_\infty$.

Lemma 3 (Autocorrelation). $\|g - g'\|_\infty \leq (\|h\|_2 + \|h'\|_2) \|h - h'\|_2$.

Lemma 4 (Jackson). On each mesh interval of length δ , if $e = h - I_\delta h$ vanishes at the endpoints, the sine series gives $\|e\|_2 \leq (\delta^2/\pi^2) \|h''\|_2$ and $\|e'\|_2 \leq (\delta/\pi) \|h''\|_2$.

Lemma 5 (Arch increment, enclosure). Write $N(w) = e^{-w/2}/(1 - e^{-2w})$ and $B(w) = e^{-2w}/(1 - e^{-2w})$. Then

$$|A(g) - A(g')| \leq (\gamma + \log \pi) |\Delta g(0)| + 2 \int_0^{2L} \min(2 \|\Delta g\|_\infty, w \operatorname{Lip}(\Delta g)) N(w) dw + 2 |\Delta g(0)| \left(\int_0^{2L} |B - N| + \int_{2L}^\infty B \right),$$

with $\int_0^\eta w N$ summed from the $w = 0$ endpoint $\sum_k \alpha_k^{-2}$ and $\int_a^b N$ by a geometric Bose series with $a > 0$. The bound is sealed, not sharp.

The r471 certificate $Q_L \succeq \mu I$ in coefficient Euclidean norm converts to an L^2 -Rayleigh floor $R_{\text{grid}} = \mu/\delta$ on the step family. The combination implication (PROVED) is: if $R_{\text{grid}} > \omega_L(\delta, \sigma_2)$ then $Q_W(h) \geq (R_{\text{grid}} - \omega) \|h\|_2^2$ for every H^2 function with that σ_2 whose interpolant lies in the certified family.

3 Sealed balance

Probe `modulus_upgrade_probe.py`, SPEC_SHA 45db1c92384913e8. Smoke 11/11 in 0.02 s; full 21/21 in 0.13 s. Every row is **SHORT**.

L	δ	R_{grid}	ω_L	R/ω	verdict
0.30 (YB)	0.0375	$8.33 \cdot 10^{-3}$	$1.40 \cdot 10^{-1}$	0.059	SHORT
0.80	0.0667	$1.98 \cdot 10^{-3}$	$1.04 \cdot 10^{-1}$	0.019	SHORT
$L_5 = 0.866$	0.108	$4.34 \cdot 10^{-3}$	$2.48 \cdot 10^{-1}$	0.017	SHORT
1.0	0.100	$3.26 \cdot 10^{-3}$	$1.80 \cdot 10^{-1}$	0.018	SHORT
1.2	0.100	$2.17 \cdot 10^{-3}$	$1.48 \cdot 10^{-1}$	0.015	SHORT
$L_8 = 1.386$	0.099	$1.38 \cdot 10^{-3}$	$1.24 \cdot 10^{-1}$	0.011	SHORT
1.4	0.100	$1.57 \cdot 10^{-3}$	$1.25 \cdot 10^{-1}$	0.013	SHORT
1.6	0.100	$9.84 \cdot 10^{-4}$	$1.13 \cdot 10^{-1}$	0.009	SHORT
1.8	0.1125	$7.80 \cdot 10^{-4}$	$1.31 \cdot 10^{-1}$	0.006	SHORT
2.0	0.125	$8.42 \cdot 10^{-4}$	$1.55 \cdot 10^{-1}$	0.005	SHORT
$L_{12} = 2.079$	0.130	$9.70 \cdot 10^{-4}$	$1.65 \cdot 10^{-1}$	0.006	SHORT
$L_{16} = 2.773$	0.154	$4.47 \cdot 10^{-4}$	$2.33 \cdot 10^{-1}$	0.002	SHORT

4 YB diagnosis

At $L = 0.3$ the prime channel is identically zero. The remaining $A + \Pi$ bound is still $17\times$ the grid floor (0.140 against 0.0083). The kill “YB upgrade must pass” therefore fires as designed: the modulus is too crude, not the grid certificate. The leading slack is the archimedean $1/w$ kernel acting on the first-order cross term $2B(I_\delta h, h - I_\delta h)$. Closing F2b needs either a cancellation estimate for that cross term or a Fourier-symbol lower bound for $A + \Pi$, not a finer mesh of the same bound.

5 What this does not do

It does not prove $\lambda_*(L) \geq 0$ at any L , including 0.3 and 2.77. It does not restate a cofinal quantifier. No RH claim.